



INVENTWOOD

# SUPERWOOD for Data Centers

Companion to the SUPERMILLS Investor Overview

September 2026

## THE THESIS

We help data centers decarbonize and improve their impact on communities, while they accelerate SUPERWOOD's journey from premium skins into *structural* applications

### 01

**The buyer is short of steel and already building with wood**

Data centers are among the fastest-growing buyers of structural steel. Customers report backlogs and shortages, and Microsoft and Meta already build data-center structures in mass timber.

### 02

**We turbocharge the wood they already use**

Mass timber replaces concrete floors. SUPERWOOD, stronger than A36 steel in tension at one-sixth the weight, adds the steel: members, skins and screens — and strengthens the mass timber itself. It ships today from SuperMill One; truss design and mass-timber enhancement are under way.

### 03

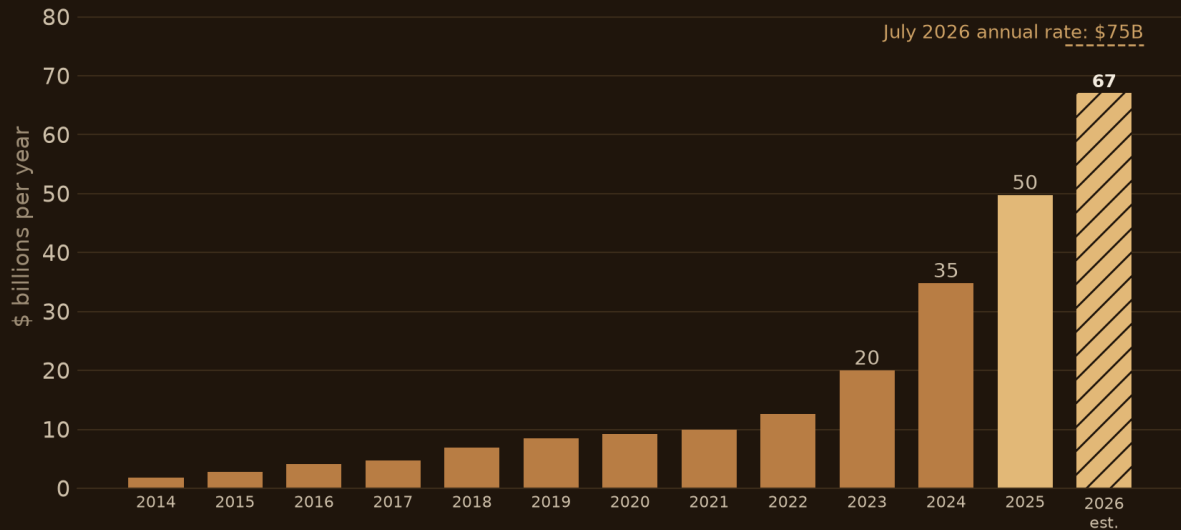
**One basis-of-design win is SuperMill Two-scale demand**

A gigawatt data center's skins alone are one to three years of SuperMill One's output; its structure is a year or more of SuperMill Two. Long term: prefabricated envelopes and, eventually, foundations.

*Note 1*

# Data center construction is growing dramatically

US data center construction is running at *\$75 billion a year*, up 57% in twelve months.



## HOW IT IS PROJECTED TO GROW

### 66 GW under construction in North America

JLL, midyear 2026 — more electricity than Germany uses. At the published 50–100 tons of structural steel per MW, that pipeline alone is 3.3–6.6 million tons of structural steel.

### Global capacity roughly triples by 2030

McKinsey: 82 GW in 2025 to about 219 GW in 2030, AI workloads driving 71% of the growth. JLL: 103 GW to 200 GW.

### Global data center capex passes \$1 trillion in 2026

Dell'Oro: up 57% in 2025, above \$1 trillion in 2026, \$1.7 trillion by 2030 — mostly servers and chips; the buildings are the Census share on the left.

# Resulting in supply chain challenges

## Power

Power transformers average 128 weeks to deliver, substation transformers more than 160. Grid connection sets the schedule.

## Aluminum

On the federal Critical Minerals List since 2022, confirmed in the 2025 list. The metal of skins, louvers and busway.

## Structural steel

Customers report backlogs and shortages. Section 232 tariffs at 50% since 2025, on full product value since April 2026.

Every week of lead time is a week of *unearned power*.

# And communities are pushing back

Community pushback now blocks or delays more data center projects than ever.



## COMPLAINT

### Aesthetics

Windowless boxes on farmland. Counties answer with setbacks and design standards.

**SUPERWOOD** Facades, fences and screens that read as architecture.

## 75 projects, about \$130 billion, blocked or delayed in one quarter

Q1 2026, as much as all of 2025. 833 opposition groups in 49 states.



## COMPLAINT

### Noise

Fans, chillers, generator tests. Property-line decibel limits are spreading.

**SUPERWOOD** Acoustic screens and barriers; quieter fan enclosures and ductwork.

## Rules are following

Henry County, Virginia: 1,000 ft setbacks, 50 dBA at the property line.

*Concept renderings*

We help with these two. *We can go much further*: into the structure of the building and what is inside it.

*Note 4*

Hyperscalers are already building structures with wood. We *turbocharge* wood.



Two Northern Virginia datacenters with cross-laminated timber floors on a steel frame — about 35% less embodied carbon than conventional steel construction, 65% less than precast. Gensler; Thornton Tomasetti.

Microsoft Source, Nov 2024



Mass-timber data-center administrative buildings: Aiken, South Carolina completed 2025 (DPR, SmartLam); Cheyenne and Montgomery under way; about 41% less embodied carbon in the materials substituted.

Meta Sustainability, 31 Jul 2025

MASS TIMBER TODAY (CLT, GLULAM)



WITH SUPERWOOD



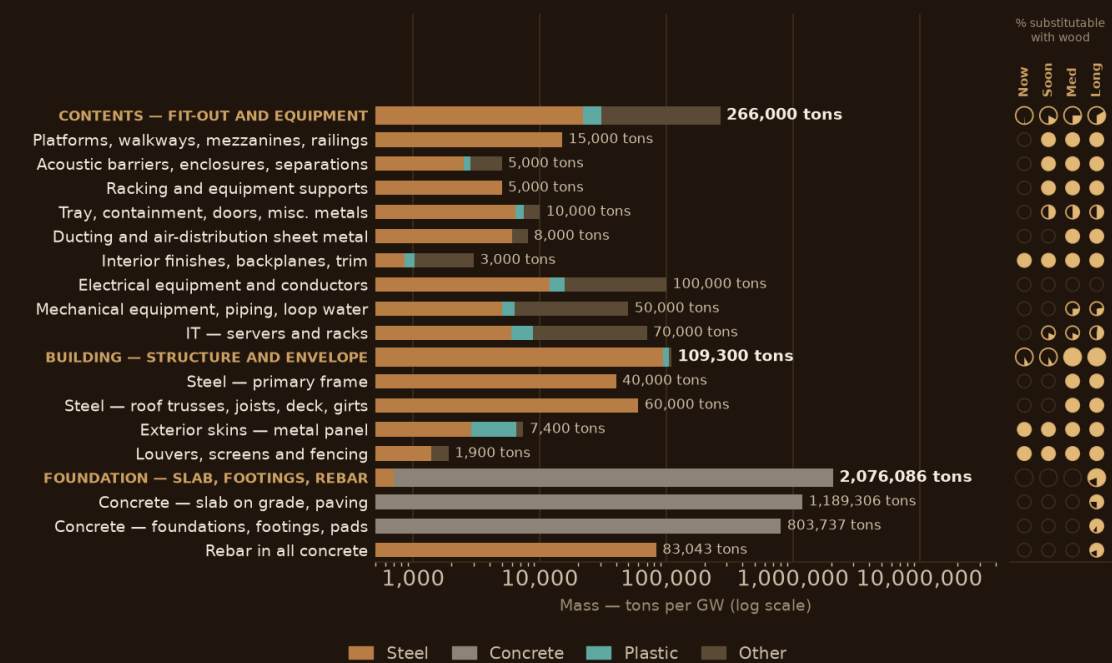
|                       |                                                        |                                                                                                                                |
|-----------------------|--------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| What it replaces      | Concrete floor slabs and decks; some columns and beams | Adds the steel: members, skins, screens and fences; later, enclosures and foundations                                          |
| Strength              | Lumber-grade; large sections carry the load            | Tensile strength above ASTM A36 steel in samples, at one-sixth the weight; thin, dense members                                 |
| Enhancing mass timber | Glulam and CLT at lumber stiffness                     | Mass-timber enhancement: SUPERWOOD outer laminations (~10% of section) raise a glulam beam's stiffness ~75% and strength ~100% |
| Exterior use          | Needs protection from weather                          | Exterior grade; resists moisture, pests and rot                                                                                |
| Fire                  | Chars; established assemblies                          | Chars; far better than ordinary wood — Class A demonstrated in testing                                                         |
| Code path             | Established in the building codes                      | Certifiable under wood standards today; SUPERWOOD-specific standards are the goal                                              |

Note 5

MATERIALS

50–80% of a data center’s building and contents are steel

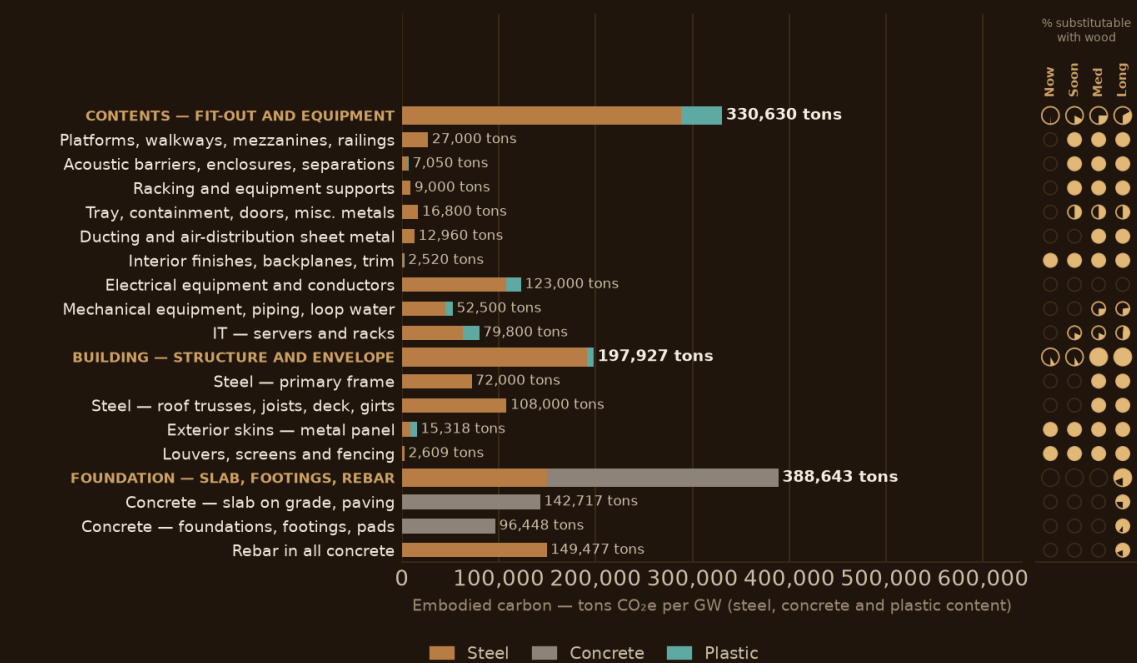
BY MASS



50–80% of the above-ground portion of a data center is steel

Structure, envelope and contents — frame, roof, skins, platforms, racks, enclosures, equipment. Skins, screens and fences now; racks, platforms and barriers next (racks are a few months of development away); frame and roofs as structure; server enclosures eventually; never the electronics.

BY EMBODIED CARBON



By embodied carbon, steel is about two-thirds of everything estimated, including the steel inside the equipment

Building materials alone are about 60% steel at the global-average factor, about a third with recycled steel. The steel and plastics in electrical, mechanical and IT equipment add roughly 250,000 tons CO<sub>2</sub>e.

Note 6

# Just one data center can consume the *entire output* of a SUPERMILL

SUPERWOOD a 1 GW data center could require, by horizon, against what each mill makes in a year.



Note 7

# SUPERWOOD's properties are a great fit for data centers



**Stronger than steel.  
One-sixth the weight.  
Made from wood, in  
America.**



## **STRONG**

Able to carry the required loads — tensile strength above ASTM A36 steel in production samples



## **LIGHT**

One-sixth the weight of steel: faster and safer to transport and install, and reduced structural requirements



## **NATURALLY BEAUTIFUL**

The texture, colors and warmth of wood — the biophilic, community-facing surface



## **DURABLE AND IMPACT RESISTANT**

Harder and more dent-resistant than oak; impact and storm resistant; exterior grade resists moisture, pests and rot



## **INSULATING, NOT CONDUCTING**

Thermal and electrical insulator — no thermal bridging, and non-conductive around electrical infrastructure



## **SOUND AND VIBRATION DAMPING**

Particularly for HVAC enclosures, acoustic barriers and equipment screens



## **RF TRANSPARENT**

Useful in niches: antenna screening, timing-antenna enclosures, telecom shelters

# Enabling data centers that are faster to build, lighter, reconfigurable and relocatable

01

## One-sixth the weight of steel

Smaller cranes and crews, easier transport, lighter foundations. Time to power is what hyperscalers optimize.

02

## Prefabricated to precision

Panels and modules built off site and set like mass timber — standardized designs, plug-and-play systems.

03

## Designed for assembly and disassembly

Reusable, reconfigurable components for infrastructure that changes with every hardware generation.

04

## Mass timber's schedule record

About 20% average schedule savings across seven case studies, up to 25–30% in others. SUPERWOOD builds the same way.



*Concept rendering*

In a fast-changing industry, this really matters.

# Skins, screens and fences ship today from SuperMill One

Every item below ships today as boards up to 8" × 16' × 3/8", exterior or interior grade. SuperMill One makes about one million square feet a year across all markets.

**IMMEDIATE** shipping from SuperMill One · **SOON** 1–3 years; straightforward applications engineering · **MEDIUM TERM** complex applications engineering such as full building systems, new form factors, or materials engineering · **LONG TERM** 5+ years; technical potential with no design or code pathway yet



Facades, cladding & rain screens



Biophilic interiors for office space



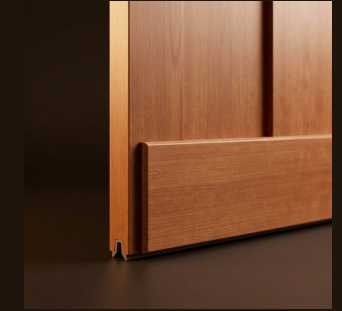
Louvers & equipment screening



Fencing for staff outdoor spaces



Security fencing around transformers & outdoor infrastructure



Trim, door kicks & sub-framing

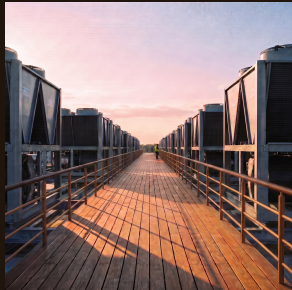
*Concept renderings*

**Why these first.** They need no assembly fire rating — where a finish rating applies, Class A has been demonstrated and is specified per order. They sit where the community and the workforce see the data center. And they are the near-term projects our hyperscaler customers asked for: facades, biophilic office interiors, staff-area fencing, security fencing around critical outdoor infrastructure.

# Other products require only straightforward engineering



Fan enclosures & ducting



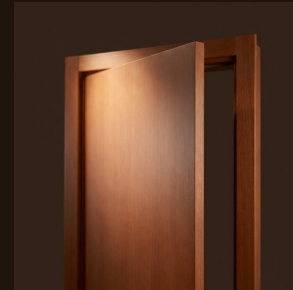
Walkways & platforms



Railings



Window mullions



Interior doors & door  
protection



Equipment backplanes



Racks & equipment  
supports

*Concept renderings*

*Each needs one scoped test program, which can be co-funded with a customer. Order depends on which a customer picks up first.*

# We're starting structural product development today, enabling scale with SUPERMILL TWO

## UNDER DEVELOPMENT WITH SUPERMILL ONE OUTPUT

Truss design is starting. Mass-timber enhancement — SUPERWOOD laminations that stiffen and strengthen glulam and CLT-type members — is under way. Light structural components (non-life-safety): strut-type supports, window mullions, deck boards.

We expect structural sales on the way to SuperMill Two. Replacing structural steel is the first thing new data-center customers ask about.

## AT SUPERMILL TWO SCALE



### Window mullions

Slender, stiff curtain-wall and storefront mullions in place of aluminum and steel — light structural, non-life-safety



### CLT-type floor, roof & wall assemblies

Starting with SUPERWOOD hybrid glulam: SUPERWOOD outer laminations on a glulam core. Thin, far stronger panels follow through the mass-timber product route



### Trusses, roofs & long-span members

Roof structure first; long-span floors and canopies follow



### Heavy equipment supports & server enclosures

Racks come soon, a few months of development; enclosures for servers and equipment follow; the electronics are never replaced

*Concept renderings*

**How qualification works.** Every structural application is qualified through a scoped test program with the customer who picks it up first, on the pathway mass timber has already opened in the building codes, with insurer acceptance for structural use. Test methods and property data are available on request.

## Longer term, we can do fully prefabricated envelopes and even fast, light foundations

### Prefabricated building envelopes

Hyperscalers are asking for this now, to simplify design and delivery and accelerate schedules.

Structure and skin shipped as panels and modules — the co-development target with our hyperscaler customers' architects and engineers. Shells that carry their own loads, weigh far less than steel and precast, and can be disassembled and moved.

### Foundations, slabs and paving

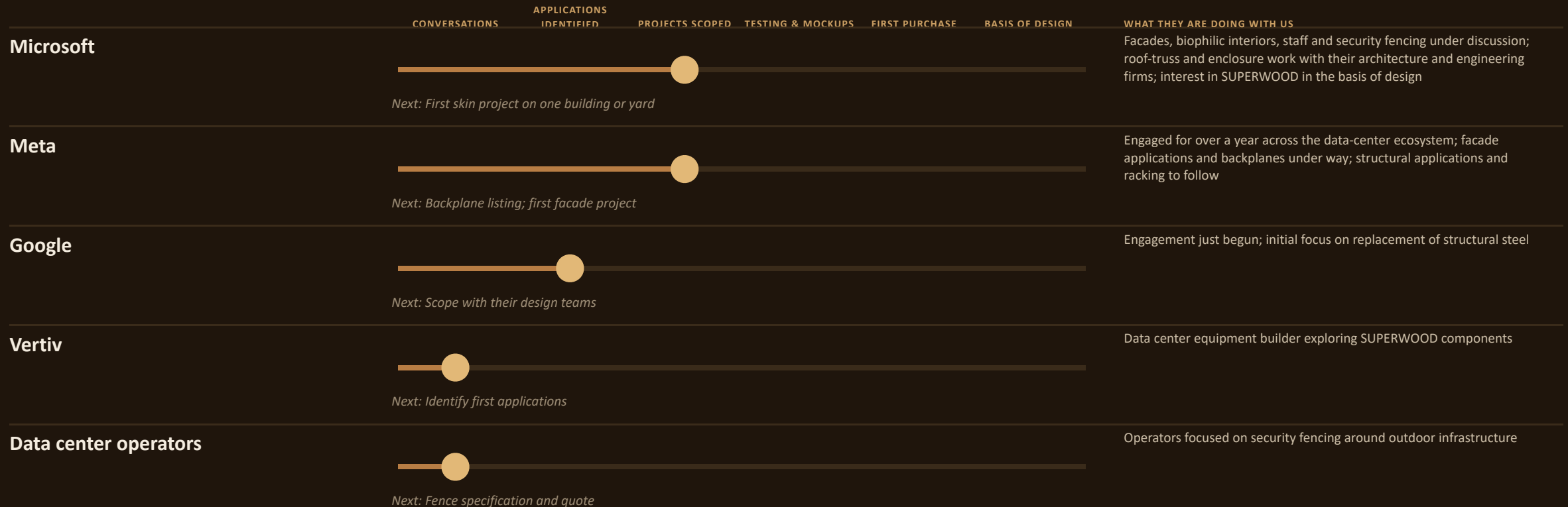
Biggest unaddressed impact both to speed of development and construction, and with huge carbon impact. Geotechnical requirements greatly impact schedule and site selection.

Lightweight, insulated SUPERWOOD foundations and slabs in place of concrete — installed faster, with lower geotechnical demands, and potentially movable.

*Stated technical potential, not an engineered plan: slab, paving and foundations are on the order of 1–2 million tons of concrete and rebar per gigawatt. No design or code pathway exists yet.*

## CUSTOMERS

# We're engaging thoughtfully and deeply with customers



ENGINEERING AND CONSTRUCTION  
COLLABORATORS

**Fast+Epp**  
Top timber engineers

**HITT**  
#1 US data center contractor, ENR 2025

**Turner**  
#1 US contractor, ENR Top 400

**Don Davies, advisor**  
Former president of MKA; structural engineer for several hyperscalers

*Note 10*

# Working on both immediate and large-scale, strategic opportunities

## Microsoft *broad engagement*

Two Northern Virginia datacenters built with cross-laminated timber floor panels on a steel frame — an estimated 35% embodied-carbon reduction vs. conventional steel construction and 65% vs. precast concrete. Design by Gensler; structural engineering by Thornton Tomasetti.

## Meta *engaged for over a year*

Piloting mass timber for data center administrative buildings: first completed in 2025 at Aiken, South Carolina (DPR, SmartLam); projects under way in Cheyenne, Wyoming and Montgomery, Alabama — about a 41% reduction in the embodied carbon of the materials substituted.

### WORKING WITH INVENTWOOD

- **Strategic interest**

Incorporating SUPERWOOD into the basis of design for future data centers

- **Application areas**

Roof trusses, leading to roofs and fast-to-build enclosures — with their principal architecture and engineering firms

- **Immediate opportunities**

Building facades · biophilic interiors · staff-area fencing · security fencing around transformers



*Concept rendering — security fencing around a transformer yard*

### WORKING WITH INVENTWOOD

- **Broad engagement**

Across every part of the data-center ecosystem, for more than a year

- **Under way**

Facade applications and equipment backplanes

- **Next**

Structural applications and racking over time



*Concept rendering — equipment backplane in an electrical room*

# We'll build on the data center opportunity to *springboard* us into much larger markets

01

## Data centers

Skins, fences and screens now; structure with SuperMill Two. A hyperscaler basis of design is the proof point.

02

## Interior and exterior products

The same boards across commercial construction. \$100B+ market.

03

## Structural members

Members proven in data centers move to offices, warehouses and industrial buildings. \$400B+ market.

04

## Optimized buildings

Prefabricated envelopes and light foundations designed around the material. \$700B+ market.

Behind all of it: steel is a \$1.5 trillion a year market and concrete \$1 trillion.

# Let's build what's *next*.

- Data centers are short of steel and already build with wood.
- SUPERWOOD turbocharges that wood: skins and fences now, structure in the medium term.
- One basis-of-design win with a hyperscaler is the demand case for SuperMill Two.

Company, mills, cost roadmap and the raise: SUPERMILLS Investor Overview

**I  NVENTWOOD**

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# Embodied carbon: LCA based on SUPERMILL TWO projections

KG CO<sub>2</sub>E, CRADLE TO GATE · EMISSIONS RIGHT OF ZERO, STORAGE LEFT

Steel — global average (BF-BOF), per kg of steel



Steel — recycled (EAF), per kg of steel



SUPERWOOD — per kg of steel substituted at 3:1 to 4:1 replacement by weight, projected: emissions right, carbon storage left



SUPERWOOD carbon estimates are based on the LCA developed by Professor Ming Hu, University of Notre Dame.

**Biogenic carbon, reported separately.** SUPERWOOD stores 0.5–1.5 kg CO<sub>2</sub>e per kg of SUPERWOOD as biogenic carbon, 0.1–0.5 kg per kg of steel it substitutes. Under EN 15804 it is released again in the end-of-life module, so it nets to zero over the life cycle; the lasting advantage is lower manufacturing emissions.

Note 12

# Notes and sources

## 1 Slide 2 · We help data centers decarbonize and improve their impact on communities, while they accelerate SUPERWOOD's journey from premium skins into structural applications

The company, team, mills, cost roadmap and the raise are in the SUPERMILLS Investor Overview. This deck covers one application. Strength: company test data vs ASTM A36; data center sizing: company estimate (slides 5–6).

## 2 Slide 3 · Data center construction is growing dramatically

US Census Bureau, Construction Spending (C30), private data center construction put in place: July 2026 preliminary seasonally adjusted annual rate \$75.2B vs \$47.8B in July 2025 (+57%); bars are calendar-year totals, not seasonally adjusted; 2026 estimated from the January to July pace; buildings only, servers excluded. JLL North America Data Center Report Midyear 2026 (66 GW). McKinsey, AI power (2025). JLL Global Data Center Outlook 2026. Dell'Oro Group (2026). Steel intensity: arXiv 2509.21312, secondary; tonnage derived. Shortage statements are what customers report to InventWood.

## 3 Slide 4 · Resulting in supply chain challenges

Lead times: Wood Mackenzie 2026 survey as reported by POWER and Data Center Knowledge. Aluminum: Federal Register, Final 2025 List of Critical Minerals (Nov 2025). Steel shortage: what customers report to InventWood. Tariffs: Congressional Research Service IN12519; Construction Dive on the full-value rule, 2026.

## 4 Slide 5 · And communities are pushing back

Data Center Watch, Q1 2026 report (June 2026). Virginia HB 153; Henry County, VA ordinance, Aug 2026. Electricity rates and water are the other leading objections; SUPERWOOD does not address them.

## 5 Slide 6 · Hyperscalers are already building structures with wood. We turbocharge wood.

Sources: news.microsoft.com, Nov 2024; Thornton Tomasetti project page; sustainability.atmeta.com, 31 Jul 2025. Logos and photographs are the companies' own, from the cited publications, used to identify the published projects. SUPERWOOD strength: company test data, parallel-to-grain tension. Hybrid-beam gains: derived from beam theory with SUPERWOOD modulus and strength, engineering write-up pending. Beams: concept renderings.

## 6 Slide 7 · 50–80% of a data center's building and contents are steel

Company estimate for a 1 GW IT-load data center, high case. Only the concrete and structural-steel intensities are published (arXiv 2509.21312, a secondary source); other rows are estimates from unit masses. Carbon factors: steel 1.8 kg CO<sub>2</sub>e/kg (global average; recycled 0.4–0.7), concrete 0.12; carbon is valued on each row's steel, concrete and plastic content (material split per component: estimates, analyses/material\_split.json; polymers at an average 3.0 kg CO<sub>2</sub>e/kg); copper, aluminum, electronics, water, gypsum and wood not valued. Concrete: footprint bottom-up with moderate soils (slab, footings, pads, paving, yard mats; the published 500–1,000 m<sup>3</sup>/MW intensity is the upper comparison). Steel share: company estimate; the 50–80% band covers metal-panel and tilt-up concrete wall designs. Horizons: immediate = shipping; soon = 1–3 years, straightforward applications engineering; medium term = complex applications engineering, new form factors or materials engineering; long term = 5+ years, technical potential with no design or code pathway yet.

## 7 Slide 8 · Just one data center can consume the entire output of a SUPERMILL

Company estimate from the slide 6 model, low–high scenarios, moderate soils; 0.3–0.6 substitution factor for steel, to be engineering-stamped per element; long-term row also includes server enclosures at 40% of IT mass (a small share of the total), electronics excluded. Mill output at 0.87 kg/sf, 1.3 t/m<sup>3</sup>. Pricing is set per application; see the SUPERMILLS Investor Overview.

## 8 Slide 9 · SUPERWOOD's properties are a great fit for data centers

Company test data; property data package and test methods available on request. Fire: chars rather than burns, far better than ordinary wood; ASTM E84 Class A demonstrated in testing, specified per order.

## 9 Slide 10 · Enabling data centers that are faster to build, lighter, reconfigurable and relocatable

Schedule figures are mass-timber case studies (ULI Urban Land: ~20% average across seven case studies, 12.7 vs 15.4 months; other studies 25–30%), used as an analogy for a material that builds the same way. No SUPERWOOD schedule data yet.

## 10 Slide 15 · We're engaging thoughtfully and deeply with customers

Next steps undated. Published backup for Microsoft and Meta on the next slide.

## 11 Slide 17 · We'll build on the data center opportunity to springboard us into much larger markets

Market sizes are company estimates from the SUPERMILLS Investor Overview; steel and concrete figures are global annual market estimates (ready-mix concrete only).

## 12 Slide 19 · Embodied carbon: LCA based on SUPERMILL TWO projections

SUPERWOOD 0.5 kg CO<sub>2</sub>e/kg manufactured and 0.5–1.5 kg/kg biogenic storage: company projections, pre-LCA. Steel 1.8 kg/kg: global BF-BOF average. EAF 0.4–0.7 kg/kg: typical published range for recycled-scrap steel. Replacement 3:1 to 4:1 by weight (0.25–0.33 kg SUPERWOOD per kg steel): company assumption. Reductions are arithmetic on these inputs.